

MHIOS verification and readiness

Component: MathGov Human Interface and Orchestration Standard v2.1 **Build:**

MG-MHIOS-2.1-20260928-CORE13.0 **Governing dependency:** MG-RL-13.0-20260926-RELEASE-I

1. Release determination

The supplied MHIOS v2.0 source has been updated to the exact attached Core 15. The result is a publication-ready **research-and-teaching companion specification, bounded record-consistency implementation, preparation workbook and synthetic test package**. It does not create a sixteenth Core document, confer execution permission, establish M2/M3 conformance for a deployed system, or validate the framework empirically.

All fifteen governing Core files are included unchanged in references/Core_15/. Their SHA-256 hashes match both the current individual uploads and the Release-I publication archive. The new MHIOS edition does not increment Canon 13.0, SGP 8.8, Aligners 5.9 or any other Core version.

2. Executed checks

Surface	Executed result	Interpretation
Python tests	133 passed	Schema, semantic, adversarial, source-reference, temporal, serialization and release-tool cases. Includes nine package-integrity/link negative tests.
Completed examples	6 consistent at the declared replay time	Synthetic decisive, authority-selected non-decisive, blocked, unassessed, recovery and scoped-authorization examples. None authorizes an action.
Governing Core identity	15 of 15 matching SHA-256 values	Exact delivered-source preservation, not a new full Core audit.
Core Word containers	14 opened through python-docx	Genuine readable Office containers; Core render evidence is not regenerated here.
Main Word/Markdown parity	Complete body and table token sequence matches	Checks the complete semantic text, excluding front cover/navigation and Markdown punctuation.
Main navigation	115 of 115 internal links resolve to their rendered heading	Unique Word bookmark names and matching PDF destinations; no live PAGeref fields.
Main PDF	41 pages rendered and visually reviewed	No blank pages or text blocks outside page boundaries. Five final typography/list-repair pages were re-rendered and reviewed; other pages were unchanged at 72 dpi.
Audit report	5 pages rendered and visually reviewed	Complete audit, source boundaries and exact repair locations.
Preparation workbook	14 sheets; 20 formulas; no cached error cells	Untouched qualification rows display UNASSESSED, not assessed failure.
Workbook behavior probes	8 of 8 matched	Untouched, incomplete, valid tokens, rights failure, unknown rights, emergency, invalid token and missing source-reference cases.
Static website reading links	Checked by reference/check_web.py	Local files and HTML anchors only; remote sites are not fetched by this test.
Final integrity	verify_release.py checks every listed file	The SHA-256 ledger is generated after all semantic, render and workbook bytes are fixed.

verify_all.py runs ten commands: checksum verification, the full test suite, artifact/source inspection, static-reading checks and six example CLI checks. It stores the command logs and receipt **outside** the publication directory and checks that the payload is unchanged after execution. The final delivered ZIP is

extracted into a clean directory and this same suite is rerun there. The external delivery receipt names the archives and their exact hashes; do not substitute this report for those executable results.

3. Implementation behavior and its limits

The checker rejects inconsistent declarations such as ordinary ranking after failed qualification, missing or duplicated required contender comparisons, a declared joint stress omitted from the result matrix, a selected candidate that is not the nominal leader, false source-file hashes, escaped local reference paths, duplicate JSON keys, nonfinite numeric values, expired or mismatched execution bindings, a missing monitoring plan and unreconciled Agent recovery.

It **does not** independently calculate RLS or CVaR, discover every stakeholder or omitted alternative, choose the required stress family, decide whether a source is true, perform an SGP evaluation, authenticate a real operator, verify a real command signature, execute a tool, or validate physical-control effectiveness. A declared complete stress matrix is still subject to substantive reviewer challenge. A matching hash proves byte identity only.

An active synthetic authorization example is evaluated at 2026-09-28T12:00:00Z. Omitting the replay clock uses present time; an old example may expire. Historical replay cannot restore live permission. The blank YAML draft intentionally fails completed-record validation.

The sixteen inherited conformance vectors are all listed in CONFORMANCE_COVERAGE.csv, which distinguishes automated checks from document-level and manual implementation tests. Describing a manual obligation is not a passed human-interaction, accessibility or operational test.

4. Preservation and migration

The 24-section organizing structure and ten inherited tables remain. Necessary source, state and interface requirements are revised in place. Existing table widths, borders, shading and style definitions are retained; 124 keep-row-together flags prevent split-row fragments. Five independent numbered sequences restart at one, and two literal Markdown emphasis spans are rendered as Word italics. Version identity moves to Appendix RELEASE; the running header/footer no longer names the old edition. Page/word metadata uses the documented render/text method.

The MHIOS preparation workbook retains its fourteen sheets and visual design. Its twenty local preparation formulas were corrected; it is **not** the unchanged Core Aligners workbook. TOKENS_QUALIFY denotes only accepted declared tokens and source-reference presence, not valid evidence, a Canon gate verdict or permission.

Existing v2.0 session records are not automatically wire-compatible with the stricter v2.1 schema. Migration creates a new linked record, supplies per-option status and reference fields, preserves unresolved values and requires revalidation. Never manufacture missing assessments merely to make an old record pass.

5. Unexecuted or unestablished surfaces

No native Excel/Calc spreadsheet recalculation or interoperability certification; no empirical rater study or cross-domain calibration; no interactive browser, keyboard, assistive-technology or participant study; no live website deployment or hosted CI execution; no actual domain authority, physical safety, cryptographic command acceptance or production-runtime test. Full external Canon schemas/registries, normative kernel libraries and applicable RPAP/PFAP remain separate dependencies. Their absence is disclosed rather than filled with invented files.

The right next use is publication of this bounded companion, independent review and implementation-specific testing. A deployment must separately demonstrate the human, organizational, legal, technical and consequence-domain conditions required by the unchanged Core.